Marco Antonio Peña Ibarra

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Education

The University of Texas Rio Grande Valley — Texas, USA

M.S. in Mathematics — Expected May 2028

B.S. in Mathematics, Minor in Computer Science — May 2026

GPA: **3.64/4.00** | *Cum Laude*

Technical Skills

Programming: Python, JavaScript, SQL, C++, R, HTML, CSS

Databases & Development: PostgreSQL, Supabase, Git, GitHub, Docker, Linux

Scientific Computing & Tools: NumPy, MATLAB, SAS, LaTeX

Technical Projects

Seminary Academic Management Portal — Live Website | GitHub

JavaScript, PostgreSQL, Supabase, Python, Docker

- Developed an academic management portal supporting **student, faculty, and administrative roles**, including enrollment, assignments, and grade management.
- Implemented secure authentication and role-based access using **Supabase Auth** and **PostgreSQL Row Level Security (RLS)**.
- Built automated SQL regression tests with **Python and Docker** to validate data isolation, permissions, and database constraints.

Coffee Origin Classification — GitHub

Python, NumPy, Logistic Regression

- Collaborated on a machine learning project to classify coffee origins from brewing time-series data using engineered numerical features.
- Implemented **logistic regression with gradient descent** and compared linear and polynomial-feature models through validation.

Parish Website – Nuestra Señora de Guadalupe — Live Website | GitHub

HTML, CSS, Responsive Web Design

- Developed a responsive public website presenting schedules, services, and parish information across desktop and mobile devices.

Professional Experience

Graduate Teaching Assistant — 2026–Present

The University of Texas Rio Grande Valley

- Support undergraduate mathematics instruction through grading, examination proctoring, and student office hours.

Peer-Led Team Learning Leader & Mentor — 2023–2026

The University of Texas Rio Grande Valley

- Facilitated mathematics learning sessions, mentored peer leaders, and developed LaTeX-based instructional materials for undergraduate courses.

Research Experience

An Introduction to p-Adic Numbers — 2026

Undergraduate Mathematics Project | Advisor: Dr. Wilson Zúñiga-Galindo

Studied foundational concepts in p-adic number theory and mathematical analysis.